

Chip Designing

Chip designers across the world strive to make faster and cheaper chips that can automate part of or the entire function of mechanical devices. The arrival of embedded systems (where intelligent devices are put inside everyday objects) and ubiquitous computing (where small computing devices proliferate to such an extent that even your shoes may be used to monitor your heartbeats) has given a boost to demand for professionals in chip design.

Chip designers are basically design engineers who define the chip's architecture, create circuit designs, run simulations, supervise layout, tape out the chip to the foundry and evaluate the prototype once the chip comes back from the laboratory.

Key skills required

A bachelor's degree in electronics, communications or instrumentation is a prerequisite. Companies look for candidates with knowledge of the circuit design, VLSI design, microcontrollers, IC design, C programming, Assembly-level programming, etc.

Companies want students who have designed a chip as part of their curriculum and also tested it in silicon. Although many companies provide job-specific training, it is expected that the candidate has technical knowledge required to absorb the training.

Who is hiring?

Today, as many as 23 of the top 25 semiconductor companies in the world have a strong presence in India. Also, there are more than 200 Indian semiconductor companies.

Some of the major recruiters are Texas Instruments (TI), PMC Sierra, Alliance Semiconductor, Analog Devices, Cadence, Synopsys, Cypress Semiconductor, HCL Technologies, Intel, IBM, Micron Tech, National Semiconductor, Philips Semiconductor, Qualcomm, Sanken, Moschip, Cradle Tech, Synplicity, Wipro and Infineon.

Engineering Management

Engineering management primarily deals with optimisation of work systems. This includes improving the efficiency and effectiveness in manufacturing as well as service sectors. Engineering management programmes typically include human resource management, industrial psychology, mathematical modeling and optimisation, quality control and operations research.

The candidate needs to know both the subjects well—technology and management. The knowledge of technology falls into place when applying managerial skills. The scientific background helps understand better how to optimise resources and develop innovations in technology. This is why MBA by itself is not suited for engineering management. Rather, you need to learn management in a technical scenario.

Key skills required

Candidates who wish to pursue engineering management as profession should be able to solve challenging problems, visualise solutions using computer modeling and design tools, and apply abstract thought to problem solving. These skills become important as engineering managers are required to oversee the design of machinery, equipment, products or systems; design and assess the feasibility of new products or processes; oversee direct production, quality assurance or maintenance; and coordinate with other units such as management, financial and marketing.

Recruiters mainly look for the ability to understand, manage and develop new and changing technology. Communication skills, client management, cost control, supply chain management and resource optimisation are some of the other factors taken into consideration while hiring.

Who is hiring?

There is no dearth of jobs for an engineering management student as all technology companies require managers in areas like industrial management, quality assurance management, energy management, operation research, airline management and environmental management.

Most jobs are in IT/software services (35 per cent), while 33 per cent are in manufacturing/FMCG (fast-moving consumer goods or consumer packaged goods) and the rest are in consulting and others.

Nokia, Siemens, Videocon, Onida, Samsung, LG, Sanmina-SCI, Flextronics, Philips, Jabil and Havells are some of the major recruiters in this industry.